

RT60 Acoustic Analysis — Rhino Instructions

aeis_rt60_rhino.py · Week 2 Deliverable

BEFORE YOU BEGIN

- You need Rhino 7 or 8 with the Python scripting engine enabled
 - Download `aeis_rt60_rhino.py` from the course site
 - Have your room model open with all surfaces visible
-

STEP 1 — SET UP YOUR LAYERS

Create one layer per material type using this naming convention:

```
AEIS_<MATERIAL>
```

Examples: `AEIS_GYPSUM_BOARD` , `AEIS_CARPET_THICK` , `AEIS_CONCRETE_BARE` , `AEIS_SEATS_OCCUPIED`

The full list of valid material names is in the script's `MATERIAL_TABLE` section and on the Week 2 course page.

Rules:

- Layer names must start with `AEIS_` (uppercase)
 - Every room surface must be on one of these layers
 - Do not leave room surfaces on the default layer
-

STEP 2 — ASSIGN SURFACES TO LAYERS

Move each wall, floor, ceiling, and seating surface to the appropriate `AEIS_` layer based on its finish material.

- A surface can only be on one layer, so choose the dominant finish
 - Openings (doors, windows) are typically ignored unless they are significant glazed areas (`AEIS_GLASS`)
 - Seating areas should use `AEIS_SEATS_OCCUPIED` or `AEIS_SEATS_UPHOLSTERED`
-

STEP 3 — PLACE A SOURCE POINT (OPTIONAL)

Create a Point object and place it on a layer named `AEIS_SOURCE` . This is the sound source location for the ray trace visualization.

If you skip this step, the script will ask you to click a location when it runs.

STEP 4 — RUN THE SCRIPT

1. In Rhino, go to **Tools** → **Python Script** → **Run**
2. Navigate to and select `aeis_rt60_rhino.py`
3. The script will begin collecting geometry from your AEIS_ layers

STEP 5 — CONFIRM THE ROOM VOLUME

A dialog will appear showing the detected volume and the method used:

- **Closed solid (ACCURATE)** — your surfaces formed a sealed room; accept this value
- **Bounding box (APPROXIMATE)** — surfaces have gaps; the volume may be overestimated

If the volume looks wrong, click **No** and enter your room volume manually in cubic meters.

To convert: cubic feet ÷ 35.315 = cubic meters

STEP 6 — SELECT PROGRAM TYPE

Choose the program type that matches your room's intended use:

OPTION	TARGET RT60
speech	0.6 – 1.0 s
mixed	1.4 – 1.8 s
music_opera	1.6 – 2.0 s
music_symph	1.8 – 2.2 s
cinema	0.3 – 0.5 s

STEP 7 — REVIEW RESULTS

The script outputs results in two places:

- Rhino command line** — full octave band table (125 Hz through 4000 Hz) with Sabine and Eyring RT60 values and status for each band
- Results dialog** — summary showing RT60 at 500 Hz, room volume, and total surface area
- Use the Eyring RT60 at 500 Hz as your primary result.** The Sabine value is provided for comparison with your hand calculation worksheet.

STEP 8 — SAVE THE HTML REPORT

When prompted, click **Yes** to save an HTML report. Choose a save location on your desktop or project folder.

The report includes:

- 500 Hz summary with status badge
- Full octave band table
- Material schedule with areas and absorption coefficients
- A screenshot of your Rhino viewport

Submit the HTML file with your assignment.

TROUBLESHOOTING

Volume is much larger than expected

Your surfaces likely have gaps and the script is using the bounding box method. Override the volume manually in Step 5 using your known room volume.

"No AEIS_ layers found"

Check that your layer names start with `AEIS_` in uppercase and that objects are assigned to those layers.

A layer is skipped with a warning

The material name after `AEIS_` does not match the table. Check spelling against the material list on the course page.

RT60 is far from your hand calc

First verify the volume is correct (Step 5). If the volume matches but RT60 differs, check that your surface areas and material assignments are consistent with your hand calc worksheet.